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Ogre elements — A distinct group of plant Ty3/gypsy-like retrotransposons

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Abstract

Ogre elements are a group of LTR retrotransposons recently discovered in legume plants, where they constitute almost 40% of the genome in some species. They are exceptional in their size (reaching 25 kb) and possess several specific features, including an intron within a polyprotein-coding region, and an extra open reading frame (ORF1) encoding a protein of unknown function located upstream of the *gag* gene. Although these features make Ogres interesting for further research, identification of additional elements from a broader range of plant taxa has been complicated by the divergence of their sequences, preventing their detection using similarity-based searches. Here we report the results of structure-based computational searches for Ogre elements in available plant genomic sequences, which proved to be more efficient and revealed occurrences of Ogres in three families of dicot plants (Leguminosae, Solanaceae and Salicaceae). In addition, a representative set of 85 elements was retrieved from a model legume species *Medicago truncatula*. All identified full-length elements were used for comparative analysis, which showed that in spite of only little conservation of their nucleotide sequences, their protein domains were highly conserved, including several regions within ORF1. Further, the elements shared the same functional regions, including a primer binding site complementary to $tRNA_{arg}$, a conserved motif within a polypurine tract, and a putative intron between the *pro* and *rt/rh* coding domains. These findings, together with analysis of their phylogenetic relationship to other retrotransposons based on similarities of *rt* domains suggest that Ogre elements from different plant taxa have a common origin and thus constitute a distinct group of Ty3/gypsy retrotransposons.
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Keywords: Transposable elements; LTR retrotransposons; Computational identification; Sequence databases; Evolution; Phylogeny

1. Introduction

Long terminal repeat (LTR) retrotransposons are widespread in plants, often constituting considerable portion of their genomes (SanMiguel and Bennetzen, 1998; Kentner et al., 2003; Chantret et al., 2005). Differential amplification of LTR retrotransposons represents one of the major forces shaping plant genomes and contributes significantly to the extraordinary variation in nuclear genome sizes found among Angiosperms (Bennetzen, 2002). This impact on the genome size is due to replicative (copy-and-paste) mechanism of the retroelement transposition, involving transcription of the parental element, reverse transcription of the resulting RNA into DNA, and subsequent integration of a new element into the genome (Kumar and Bennetzen, 1999; Havecker et al., 2004).

LTR retrotransposons are distinguished from other types of retroelements by the presence of direct terminal repeats, which are crucial for transcription and replication of the elements (Kumar and Bennetzen, 1999). In the vicinity of the LTRs there are other regions important for element replication: a primer binding site (PBS) and a polypurine tract (PPT). The PBS is made up of a short sequence adjacent to the 5′ LTR and is usually complementary to the 3′ end of a certain type of transfer RNA (tRNA) which is used as a primer to initiate synthesis of a minus DNA strand. The type of tRNA is always characteristic for a particular retrotransposon family, with $tRNA_{ini}$ being the most frequent (Chavanne et al., 1998; Hull, 2001; Wilhelm and Wilhelm, 2001). The PPT is located upstream of the 3′ LTR and is a site of plus strand synthesis initiation during retrotransposon

Abbreviations: EST, expressed sequence tag; IN, integrase; LTR, long terminal repeat; ORF, open reading frame; PBS, primer binding site; PPT, polypurine tract; RH, RNaseH; RT, reverse transcriptase.

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replication (Wilhelm and Wilhelm, 2001). The proteins required for element replication and transposition are expressed as a Gag-Pro-Pol polyprotein which is encoded by *gag*, *pro* and *pol* genes located between the LTRs. Depending on the expression strategy employed by each particular family of LTR retrotransposons, these genes are usually organized into one or two ORFs (Peterson-Burch and Voytas, 2002; Gao et al., 2003; Havecker et al., 2004). The *gag* gene encodes structural proteins, which are needed mainly for an assembly of virus-like particles and RNA packaging. The *pol* gene encodes for reverse transcriptase/RNaseH (RT/RH) and integrase (IN) proteins. They ensure retroelement replication via reverse transcription and integration of a new copy into a genome, respectively. The retrotransposon genes are expressed as various types of precursor polyproteins. Individual functional proteins are released from the polyproteins by a protease (Pro), which is encoded by the *pro* gene. LTR retrotransposons are further classified either as Ty3/gypsy (Metaviridae) or Ty1/copia (Pseudoviridae) according to the domain order within the *pol* gene (Kumar and Bennetzen, 1999; Hull, 2001; Havecker et al., 2004).

While the polyprotein-coding region (*gag-pro-pol*) encodes all proteins required for element replication and integration, some LTR retrotransposons also possess extra ORFs whose functions are unclear. In some cases, the extra ORF is only a transduced cellular gene with no obvious function for the retrotransposon life cycle (Bureau et al., 1994; Jin and Bennetzen, 1994). The extra ORF in some other retrotransposons, however, might have importance for the retrotransposon life cycle because it was found in a broad spectrum of retrotransposons. Such an example is an *env*-like ORF located downstream of the polyprotein-coding region in *env*-class retrotransposons (Vicient et al., 2001; Wright and Voytas, 2002; Havecker et al., 2004). Although most extra ORFs reported so far were located downstream of the polyprotein-coding region (Havecker et al., 2004), there are also rare cases of extra ORFs located upstream (Kumekawa et al., 1999). The upstream extra ORF is one of the most characteristic features of Ogre retrotransposons, which were recently identified in several genera of legume plants (Neumann et al., 2003, 2006).

Ogre elements constitute a significant portion of genomes of some *Pisum* and *Vicia* species, and are also present in the model plant *Medicago truncatula* (Neumann et al., 2003, 2006). They occur in about 10^4 copies in *Pisum sativum*, and reach 10^5 copies in several *Vicia* species, representing up to 38% of their nuclear DNA. Thus, amplification of this single retrotransposon family increased the genome size of some of their hosts by more than 50% (Neumann et al., 2006). Such profound effects were caused not only by high copy numbers but also due to the exceptional length of Ogre elements, which can reach up to 25 kb and contain LTRs over 6 kb long. Although they can be classified as a member of the Metaviridae (Ty3/gypsy-like elements) based on the order of their protein-coding domains, Ogre elements posses several structural features that distinguish them from other Ty3/gypsy retrotransposons. In addition to the extra open reading frame (designated ORF1) of unknown origin and function these distinguishing features include an intron located between the *gag-pro* and *pol* ORFs, and a rare type of PBS complementary to the 3′ end of $tRNA_{arg}$ (Neumann et al., 2003, 2006).

Our previous experiments revealed that although Ogre sequences are highly conserved within *P. sativum* and *Vicia pannonica*, they display considerable differences between the species. This prevented reliable experimental (hybridization-based) or computational (sequence similarity-based) detection of Ogre elements in other plant taxa, which both require a certain degree of sequence conservation. However, the structure of these elements was found to be conserved; they always contain ORF1 and mostly also contain the intron region. Thus, we improved our Ogre detection strategy by primarily searching for Ogre-like elements based on their typical gene structure rather than on direct sequence similarity to known elements. Here we show that Ogre elements occur outside the legume family and explore their distribution among various plant taxa. We also take advantage of the rapidly growing sequence data from the model plant, *M. truncatula*, to identify a representative subset of Ogres from this species and to determine their variability. The whole set of Ogre sequences is then used as a basis for an overall comparative analysis to characterize Ogre structural and functional regions, further investigate ORF1, and determine the phylogenetic relationship of Ogres to other retrotransposons.

2. Materials and methods

2.1. Input data

Automated structure-based searches for Ogre elements were conducted on a dataset consisting of all green plant (Viridiplantae) nuclear DNA sequences at least 15 kb long which were available from GenBank on May 9, 2006 (downloaded from NCBI Entrez server, http://www.ncbi.nlm.nih.gov/entrez/). Genomes of *Arabidopsis thaliana* and *Oryza sativa* ssp. *japonica* were represented by five and twelve pseudomolecules, respectively, representing individual chromosomes (accession numbers NC_003070, NC_003071, NC_003074–NC_003076, AP008207–AP008218). In addition, unfinished working draft sequences (WDS) available at the time of analysis were also included in the searches. In total, 800,654,539 bp in 1772 sequences were analyzed. Distribution of these sequences among individual plant taxa is specified in Supplementary material S1.

2.2. Structure-based identification of potential Ogre elements

Identification of all types of intact LTR retrotransposons was done using the LTR_STRUC program (McCarthy and McDonald, 2003), which was run at the search intensity level 3. Since our preliminary analyses using sample datasets had revealed that the program missed some elements if they were in reverse orientation, all sequences were scanned in both orientations. The elements identified by LTR_STRUC were then further analyzed using a set of Perl scripts to identify those elements containing an extra open reading frame at least 100 bp long and located upstream of the Gag-coding region. This was achieved

by finding all ORFs of this minimum size within the entire element sequence (excluding LTRs), identifying the ORF containing the Gag-coding domain by locally implemented RPS-BLAST (Marchler-Bauer et al., 2003), and determining if there was an additional ORF upstream of this region. At the same time, detection of other protein-coding domains and extraction of other information about the elements were performed for use in further analysis. All these automated searches were executed on an IBM xSeries 226 dual processor server with 2 GB RAM running under Gentoo Linux operating system.

2.3. Similarity-based searches and further analyses

Potential Ogre elements identified in previous searches were individually examined for positions of their ORFs and for the distribution of Gag-, Pro-, RT/RH- and IN-coding domains using Staden package (Staden, 1996) and RPS-BLAST. Verified Ogre elements were employed as queries in nucleotide and protein database similarity searches using BLAST (Altschul et al., 1997). The elements were further analyzed for their mutual similarities using dotter (Sonnhammer and Durbin, 1995). Multiple sequence alignments were performed using ClustalW (Thompson et al., 1994). Plots of sequence conservation along the alignments were generated by the PLOTCON program, which is a part of the EMBOSS package (Rice et al., 2000; see also http://emboss.sourceforge.net/apps/release/4.0/emboss/apps/plotcon.html for detailed description of the algorithm used for evaluation of alignment conservation). Pairwise similarities of LTRs from *M. truncatula* elements were extracted from their ClustalW analysis and transformed into a graph as described by Macas et al. (in press). Sequences of tRNAs used for identification of PBS were obtained from the *A. thaliana* tRNA database (Lowe and Eddy, 1997; http://rna.wustl.edu). Putative intron splice sites were predicted at the NetGene2 server (Hebsgaard et al., 1996; http://www.cbs.dtu.dk/services/NetGene2/). Potential transmembrane domains were predicted using TMAP (Persson and Argos, 1994) and TMHMM (Krogh et al., 2001; http://www.cbs.dtu.dk/services/TMHMM/). Phylogenetic analysis was done using multiple sequence alignment of reverse transcriptase domains and a phylogenetic tree was calculated by neighbor-joining method (bootstrap values calculated from 1000 replicates) using the ClustalX program (Thompson et al., 1997). Graphical representation of the tree was drawn using the Phylojava client/server tool. Reverse transcriptase sequences were mostly taken from the previously published alignment ALIGN_000602 (Vicient et al., 2001).

Table 1
Full-length Ogre elements

Species	Element	Length [bp]	LTRs		TSD[a]	Sequence		Additional elements[b]	Notes
			Length [bp]	Similarity [%]		Accession no.	Position		
Leguminosae									
Pisum sativum	Ogre-PS1	22,753	5040/5071[c]	99.6	CCGTG	AY299398.1	14,501–37,253	–	Neumann et al. (2003)
Vicia pannonica	Ogre-VP1	25,049	6438	99.9	ATGCC	AY936172.1	25,540–492	–	Neumann et al. (2006)
Medicago truncatula[d]	Ogre-MT1A	18,237	3323	100.0	AAGTT	AC122728.32	36,056–54,292	11	
	Ogre-MT1B	18,752	3567	100.0	TATGT	AC146709.11	35,348–54,099	15	
	Ogre-MT1C	18,350	3298	100.0	AAACT	AC174378.9	78,195–59,846	19	
	Ogre-MT1D	18,854	3586	100.0	TTAGT	AC150445.24	49,330–68,183	13	
	Ogre-MT2	16,071	2006/2005[c]	99.9	ACAAG	AC148171.16	122,054–138,124	12	
	Ogre-MT3	16,325	2224	100.0	ATTGT	AC151524.2	76,334–92,658	4	
	Ogre-MT4	15,858	1884	100.0	CAAGA	AC145061.27	77,054–92,911	4	
Salicaceae									
Populus trichocarpa	Ogre-PT1	17,799	2218	99.8	AACCC	AC149300.1	19,339–1541	1	
	Ogre-PT2	17,832	2775/2842[c]	96.7	GTACT	AC182676.2	36,607–54,438	–	Identified by similarity searches to Ogre-PT1
	Ogre-PT3	17,076	2992/2995[c]	99.6	GTTTT	AC149300.1	92,345–109,420	–	Identified by similarity searches to Ogre-PT1
Solanaceae									
Lycopersicon esculentum	Ogre-LE1	20,272	4003/4008[c]	99.4	CCTAT	AC171735.1	62,208–82,479	–	
Solanum demissum	Ogre-SD1	18,330	2277/2287[c]	97.1	CCTGG	AC146506.1	59,796–41,467	–	

[a] Target site duplication.
[b] Number of other identified full-length elements which show high sequence similarity to the element listed in the table.
[c] LTRs differ in size.
[d] A total of 85 unique elements were identified in *M. truncatula*. There are only seven typical elements listed in this table; the complete list is provided as Supplementary material S2.

Table 2
Occurrence of Ogre sequences in various plant taxa revealed by similarity searches to nucleotide sequences of full-length elements or to protein sequences encoded by ORF1

Species	BLASTn with full-length elements[a]		tBLASTn with ORF1 proteins[a]		Notes
	nr[b]	ESTs[c]	nr[b]	ESTs[c]	
Leguminosae					
Cicer arietinum	AC145813 ($1e^{-25}$, PT2)	–	–	–	
Glycine max	–	CA820006 ($5e^{-51}$, MT2), BQ473876 ($5e^{-39}$, MT4),	AY075133 ($2e^{-31}$, PS1; $2e^{-28}$, MT1)	BI321731 ($2e^{-18}$, MT1)	
Lens culinaris	AJ401231 ($3e^{-30}$, PS1)	–	–	–	
Medicago truncatula	n.t.	Multiple hits (<$1e^{-100}$, MT1–4)	n.t.	n.t.	
Pisum sativum	AY299394–AY299397 (0.0, PS1)	–	n.t	–	Several truncated Ogre elements were described by Neumann et al. (2003)
Vicia faba	M16858 ($5e^{-32}$, PS1)	–	M16856 ($3e^{-59}$, VP1), AJ011710 ($1e^{-31}$, PS1)	–	
Vicia melanops	AJ391759 (0.0, VP1), 10 more hits (<$1e^{-100}$, VP1)	–	–	–	Fragments of Ogre sequences characterized in Neumann et al. (2006)
Vicia narbonensis	AJ391770 ($4e^{-29}$, MT3), AJ391777 ($2e^{-15}$, MT1)	–	AJ391771 ($6e^{-27}$, VP1; $5e^{-20}$, MT1)	–	Fragments of Ogre sequences characterized in Neumann et al. (2006)
Vicia pannonica	AY936172 (0.0, VP1)	–	n.t.	–	The clone AY936172 contains Ogre-VP1 and partial sequences of two additional Ogre elements (Neumann et al., 2006)
Vicia sativa	AJ851082 ($8e^{-56}$, VP1), AJ391782 ($1e^{-23}$, VP1)	–	–	–	Fragments of Ogre sequences characterized in Neumann et al. (2006)
Salicaceae					
Populus deltoides	AJ416708 ($1e^{-167}$, PT1)	–	–	–	
Populus tremula	–	BU888652 ($9e^{-161}$, PT2)	–	–	
Populus trichocarpa	AC149424 (0.0, PT1), AC182687 (0.0, PT3)	CV237498 ($2e^{-115}$, PT1)	AC149480 ($4e^{-139}$, PT1)	–	
Solanaceae					
Capsicum annuum	AY702979 ($8e^{-14}$, SD1)	CO910407 ($8e^{-72}$, SD1)	–	–	
Lycopersicon esculentum	AC122544 (0.0, LE1)	AJ784460 (0.0, LE1), 5 more hits (<$1e^{-100}$, LE1)	X87372 ($9e^{-38}$, SD1), AY850394 ($3e^{-20}$, LE1)	BP882563 ($1e^{-66}$, LE1)	AC122544 contains solo-LTR of Ogre element (2333 bp)
Lycopersicon hirsutum	AF534343 ($7e^{-101}$, LE1)	–	–	–	
Lycopersicon pennellii	–	AW399019 ($3e^{-87}$, LE1)	–	–	
Nicotiana tabacum	–	BP528973 ($4e^{-18}$, SD1)	AB032534 ($2e^{-99}$, SD1), AB186042 ($5e^{-71}$, SD1)	BP128793 ($6e^{-27}$, LE1)	
Nicotiana benthamiana	–	–	–	CK284994 ($1e^{-35}$, SD1), CK283508 ($3e^{-30}$, LE1)	
Solanum caripense	AJ716194 ($7e^{-36}$, LE1)	–	–	–	
Solanum demissum	AC151801 (0.0, SD1)	–	–	–	AC151801 contains rearranged and truncated Ogre element of about 11 kb
Solanum tuberosum	X04077 ($3e^{-19}$, SD1)	DR036818 ($2e^{-63}$, LE1), CK717884 ($3e^{-25}$, SD1)	AY370339 ($4e^{-33}$, LE1)	CK267231 ($6e^{-62}$, LE1)	
Euphorbiaceae					
Hevea brasiliensis	–	–	AY962223 ($8e^{-36}$, PT1; $5e^{-17}$, MT4; $1e^{-12}$, LE1)	–	
Euphorbia esula	–	DV139756 ($2e^{-11}$, PS1)	–	–	
Malvaceae					
Gossypium hirsutum	–	–	–	DT559130 ($3e^{-20}$, PT2; $1e^{-10}$, MT2, PT1)	

[a] Accession numbers of sequences producing the best scores; BLAST *E*-values and specifications of the elements used as a query are given in parenthesis. n.t. = not tested.
[b] Searches were performed on non-redundant nucleotide sequence databases (excluding ESTs).
[c] Searches were performed on ESTs divisions of nucleotide sequence databases.

3. Results and discussion

3.1. Computer-based screening for novel Ogre elements

The search for novel Ogre elements in available green plant (Viridiplantae) genomic sequences (Supplementary material S1) was performed in three consecutive steps. First, the sequences were scanned using the LTR_STRUC program (McCarthy and McDonald, 2003) in order to identify potential full-length LTR retrotransposons with well-preserved LTRs flanked by target site duplications. As expected, thousands of such sequences representing all types of LTR retrotransposons were found. Second, the sequences of these elements were searched for the presence of an extra open reading frame (ORF) upstream of the Gag-coding region, which is one of the characteristic features of Ogre elements. Third, the elements containing such ORFs were examined in detail in order to confirm their assignment as novel Ogre sequences. Many false-positives were rejected at this point, mostly including elements with mutations generating stop codons upstream of the *gag* domain. Finally, this procedure resulted in the identification of Ogre elements in genomic sequences from six plant species (Table 1). It successfully detected the only two full-length Ogre sequences previously described in *P. sativum* and *V. pannonica* (Ogre-PS1 and Ogre-VP1), proving its sensitivity and efficiency. In addition to these two elements, 85 new Ogre elements were found in the *M. truncatula* genome (Supplementary material S2). Novel elements were also detected in black cottonwood (*Populus trichocarpa*, family Salicaceae) and in two species of the Solanaceae family — tomato (*Lycopersicon esculentum*) and *Solanum demissum*.

3.2. Distribution and sequence variability of Ogre elements in different plant taxa

A further search for additional Ogre sequences was performed using similarity-based searching using nucleotide sequences of previously identified full-length Ogre elements or ORF1-encoded protein sequences of these elements as queries. Three additional full-length elements were identified in *P. trichocarpa*, two of which were significantly diverged from a query element (Ogre-PT1), thus representing different subfamilies (Table 1). In addition, a number of partial Ogre sequences were identified. They mostly represented short clones that could not accommodate the full-length elements, or truncated and rearranged Ogre sequences within long-insert clones (Table 2). Examples of the latter case include a rearranged element of about 11 kb in *S. demissum* sequence AC151801, and a solo-LTR (2333 bp) in *L. esculentum* sequence AC122544. It should be noted that some of the similarities were also found in EST clones, thus proving that Ogre sequences are transcribed. These observations are in agreement with our previous results showing that Ogre elements are constitutively transcribed in *P. sativum* and *V. pannonica* (Neumann et al., 2003, 2006).

In summary, partial Ogre sequences were detected in a broader range of species than full-length elements (Table 2). They were abundant in the legume family (detected in the genera of *Cicer*, *Glycine*, *Lens*, *Medicago*, *Pisum*, *Vicia*), in Solanaceae (*Capsicum*, *Lycopersicon*, *Nicotiana*, *Solanum*), and they were also identified in three *Populus* species (Salicaceae). In addition, a few similarities to sequences from Euphorbiaceae (*Hevea*, *Euphorbia*) and Malvaceae (*Gossypium*) were detected; however, further sequences will have to be identified to prove the presence of Ogre elements in these plant families.

In *M. truncatula*, 85 full-length elements were identified in the primary structure-based searches. The total number of Ogre elements (full-length and partial) in available *M. truncatula* sequences was estimated to be 130–150, based on the number of hits obtained in tBLASTn search with ORF1 sequence. Thus, we consider the retrieved full-length sequences to be a representative subset of Ogre elements for this species. Moreover, they include complete sequences with mostly intact ORFs and LTR similarities reaching up to 100%, thus likely representing recently active elements. Such a large dataset allowed the investigation of intraspecific sequence variability between these elements, which was found to be the highest in their LTRs and the lowest within the coding regions. As the LTR sequences appeared to evolve at the fastest rate, we used them for discriminating groups of possibly related elements. Graphical representation of pairwise similarities between 5′ LTRs of all 85 elements (Fig. 1) demonstrates the presence of distinct subfamilies, with only little similarity (<20%) in their LTR sequences. Based on this classification and on sequence alignments of the complete elements (not shown), we identified four basic Ogre subfamilies in *M. truncatula* (MT1–MT4), and

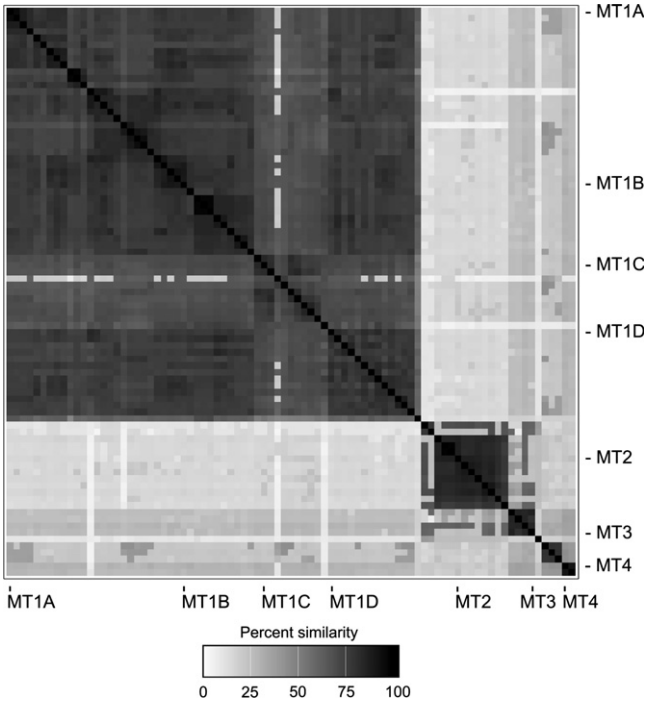


Fig. 1. Dot-plot representation of pairwise similarities between 5′ LTRs of all identified *Medicago truncatula* elements. A total of 85 sequences are arranged by subfamily assignment and their percentage similarities displayed as shades of gray corresponding to the scale below the graph. Positions of the LTRs of the elements selected for further analysis are shown.

selected one representative element from each subfamily for further analyses. Since the subfamily MT1 was still relatively heterogeneous and included most of the elements (62 out of 85), we selected four element subtypes (MT1A–MT1D) for its representation (Fig. 1 and Table 1).

Although only four complete elements were available for *P. trichocarpa*, they showed similar sequence diversification as *M. truncatula* elements, with only two highly similar sequences (Table 1). Comparison of nucleotide sequences of elements from different species revealed their further divergence, with almost no similarities found except for parts of the coding regions, with *pol* being the most conserved (Supplementary material S3). However, coding sequences appeared to be better conserved on the protein level, especially in the regions of their functional domains (Fig. 2A).

3.3. Comparative analysis of Ogre elements

Full-length Ogre elements ranged from 15,858 bp (Ogre-MT4) to 25,049 bp (Ogre-VP1) but these differences could be mainly attributed to length variability of their LTRs, which ranged from 1884 bp (Ogre-MT4) to 6438 bp (Ogre-VP1, Table 1). In spite of the considerable divergence in their nucleotide sequences, the features as the PBS, the PPT and the structure of the coding region were conserved across all Ogre elements, as depicted on Fig. 2. The PBS was located 3–7 bp downstream of the 5′ LTR and was complementary to 18–19 nt at the 3′ end of either of four types of $tRNA_{arg}$ (tRNA28, 31, 55, and 59; Lowe and Eddy, 1997). Putative PPT sequences were identified as purine-rich regions upstream of the 3′ LTR. Although these regions differed in length (12–26 bp) and sequence, they all contained a conserved motif 5′-AGGGG-3′, which started 9–10 bp upstream of the 3′ LTR. Most Ogre elements also contained an array of tandem repeats varying in total size, sequence, and length of the repeat monomers, which was located in the 3′ untranslated region upstream of the PPT. Interestingly, short arrays of tandem repeats located between *pol* and 3′ LTR occur also in several other Ty3/gypsy retrotransposons, including *Grande1* and *Cinful* in *Zea*, and *Retand* in *Silene latifolia* (Martínez-Izquierdo et al., 1997;

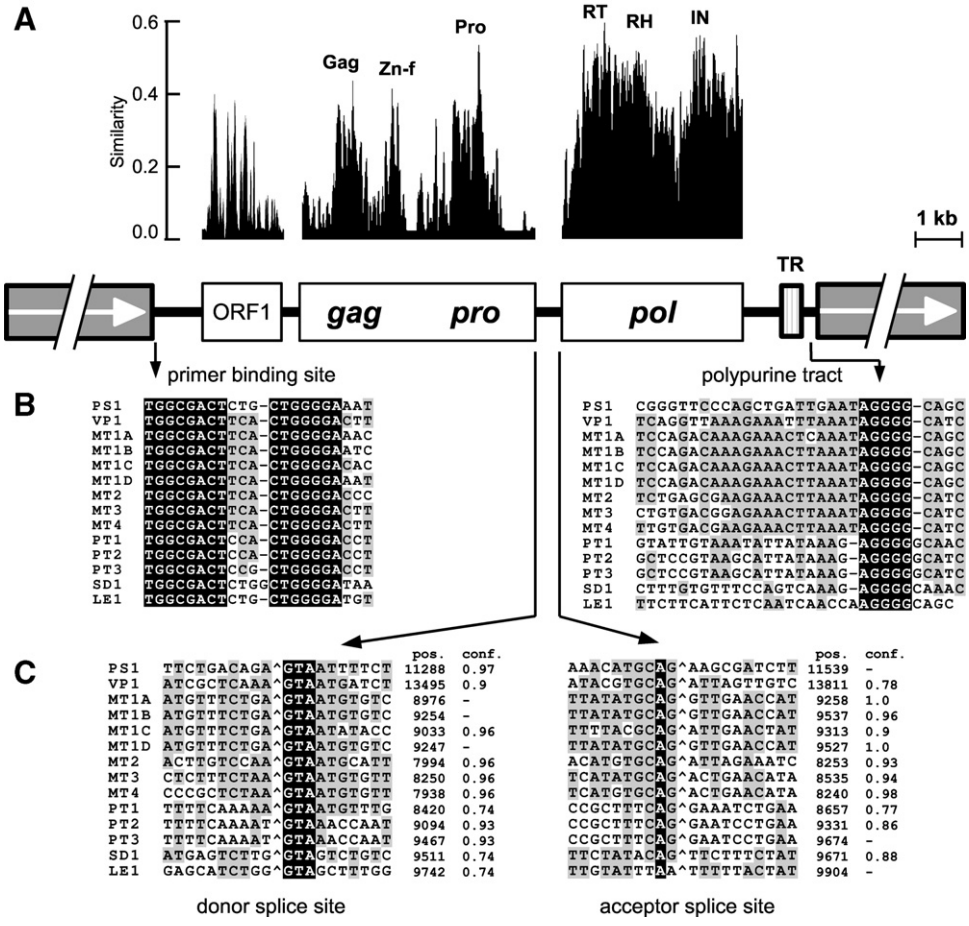


Fig. 2. Schematic representation of Ogre elements and sequence comparison of their functional and coding regions. A. A general diagram of an Ogre element displaying the relative positions of the coding regions (open boxes), the region containing tandem repeats (TR), and the LTRs (gray boxes with arrows; not drawn to scale). The graphs above the coding regions represent their sequence conservation and display average similarities among aligned protein sequences deduced from the elements listed in Table 1 (calculated using the PLOTCON program for a window of 10 residues). Positions of individual protein domains are marked by abbreviations at the top of the graph (Zn-f is a zinc-finger motif downstream of the Gag domain). B. Multiple sequence alignments of primer binding sites and polypurine tracts from the elements listed in Table 1. C. Alignments of predicted donor and acceptor splice sites of putative intron sequences (pos. — position within the element, conf. — confidence).

Sanz-Alferez et al., 2003; Kejnovský et al., 2006). However, origin and eventual function of these repeats are unknown.

The coding region was organized into three non-overlapping ORFs including ORF1, *gag-pro*, and *pol* ORFs. The *gag-pro* and *pol* ORFs are about 3.2–4.3 kbp and 3.1–3.5 kbp long, respectively, encoding all typical retrotransposon proteins (Fig. 2A). As we have previously demonstrated for Ogre-PS1 (Neumann et al., 2003), the non-coding sequence separating *gag-pro* and *pol* ORFs is an intron whose partial splicing probably regulates a ratio between Gag-Pro and Gag-Pro-Pol polyproteins. To assess if the intron occurs also in the other Ogre elements, we analyzed the element sequences listed in Table 1 using the NetGene2 server (Hebsgaard et al., 1996). Donor and acceptor sites were predicted for most Ogre elements (Fig. 2C). With the exception of Ogre-LE1 having a mutated sequence, all analyzed elements contained conserved GT and AG dinucleotides at the donor and acceptor site, respectively. However, the putative introns from different Ogre families showed high variability in length (161–317 bp) and in sequence (data not shown). Hypothetical splicing of all predicted introns results in a fusion of *gag-pro* and *pol* ORFs into one reading frame, which could be translated into a full polyprotein sequence.

The ORF1 was found to be about 1300–1600 bp long and was separated from the *gag-pol* ORF by 300–500 bp long noncoding sequences containing several stop codons. All elements listed in Table 1 contained ORF1 coding regions devoid of nonsense mutations, except for Ogre-PT2 where it was interrupted by a stop codon. Multiple alignment of protein sequences encoded by ORF1s from different elements revealed several conserved regions which may be important for protein function (Fig. 2A and Supplementary material S4). However, no known functional or structural protein domains could be associated with these regions. The only similarities to the conserved domain database (CDD, Marchler-Bauer et al., 2003) were detected for Ogre-PT1 and Ogre-SD1, each containing a chromosome segregation ATPase domain (*E*-values of $2e^{-7}$ and $2e^{-6}$, respectively), but they were located in regions which were not conserved among ORF1 proteins (Supplementary material S4). We also performed computer predictions of potential transmembrane domains that could be indicative of *env*-like retrotransposon proteins (Vicient et al., 2001; Wright and Voytas, 2002; Neumann et al., 2005). The analysis of all aligned ORF1 proteins with the TMAP program (Persson and Argos, 1994) revealed a potential transmembrane domain in one of the less conserved regions (positions 200–250 of the alignment shown on Supplementary material S4); however, examination of individual sequences using a different program, TMHMM (Krogh et al., 2001), predicted transmembrane domains only in a minority of ORF1 proteins.

3.4. Phylogenetic relationship to other LTR retrotransposons

To assess a relationship of Ogres to other plant retrotransposons, we carried out their phylogenetic analysis based on reverse transcriptase protein domains. All Ogre sequences, although from different plant families, formed well-distinguished clade within Ty3/gypsy retrotransposons (100% bootstrap value, Fig. 3) and were separated from other elements, even from those present in the same species (Ogre-PS elements were separated from other pea retrotransposons Peabody, Cyclops and PIGY). The Ogre sequences were further clustered into groups corresponding to their species of origin. These findings imply a common origin of Ogre elements and their sequence diversification during speciation of their hosts. Together with the other common structural features described

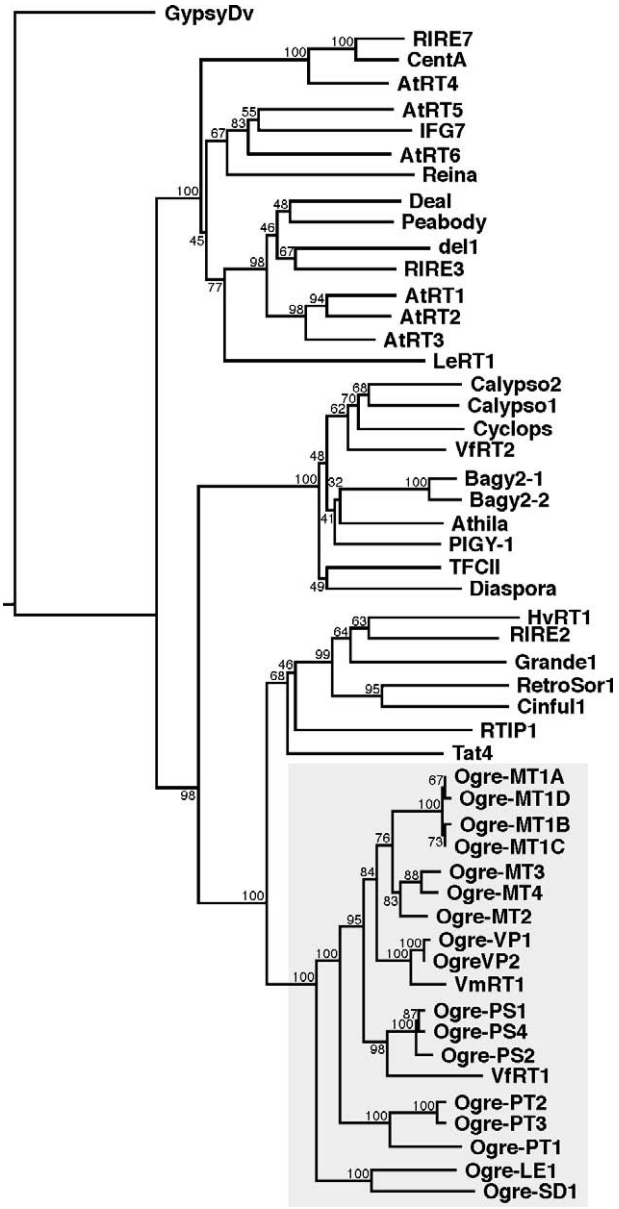


Fig. 3. A phylogenetic tree based on the analysis of the RT domains. The RT sequences extracted from Ogre retrotransposons were included in the alignment ALIGN_000602 of Vicient et al. (2001) and the phylogenetic analysis was performed using ClustalX. The positions of Ogre sequences in the tree are marked by shading and their names correspond to Table 1. Ogre-PS2 and Ogre-PS4 represent partial Ogre sequences described by Neumann et al. (2003); VfRT and VmRT sequences, which clustered together with Ogres, are partial retrotransposon sequences from *Vicia faba* and *Vicia melanops*, respectively, most likely representing fragments of Ogre sequences in these species. The numbers on the branches are bootstrap values (%) from 1000 replicates.

here our results justify considering Ogres as a distinct group of plant LTR retrotransposons.

3.5. Conclusion

Compared to our previous studies employing searches exclusively based on sequence similarity (Neumann et al., 2003, 2006), the approach used here proved to be more efficient for identifying Ogre elements and demonstrated their occurrence in several plant families outside of the legumes. On the other hand, the question of the overall distribution of Ogres in plants remains open, owing to the limited sequence information available for many plant taxa (Supplementary material S1). The absence of Ogre elements in extensively sequenced monocots, including *O. sativa*, and also in a dicot model species *A. thaliana*, suggests that the ancestral Ogre element appeared later during evolution of dicot plants, or that its descendants disappeared from some plant lineages.

Yet another question concerns the roles of ORF1 and intron sequences in Ogre expression, replication and co-evolution with its host species. The partial intron splicing, reported for pea Ogre elements (Neumann et al., 2003), can serve as a mechanism regulating the ratio of Gag-Pro to Gag-Pro-Pol proteins. However, it is not clear how the ratio of spliced to unspliced transcripts is maintained. It should also be noted that a mechanism may exist to prevent the reintegration of elements originating from the spliced transcripts back into the genome, otherwise the intron would be gradually lost from the Ogre population. It is tempting to speculate about the importance of the ORF1-encoded protein in these processes, although at present there is no supporting evidence available. Thus, further experiments should be aimed at elucidating these processes, as well as at the other interesting features of Ogre elements, including the mechanisms of translation of ORF1 and *gag-pro-pol* ORFs from polycistronic Ogre transcripts, and the structure of exceptionally long Ogre LTRs which provide regulatory regions directing the element's transcription in many plant species.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.gene.2006.08.007.

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